



LAB 3: PROPOSITIONAL LOGIC & MODEL CHECKING

**College of Engineering
Department of Computer Science
CS3081 Artificial Intelligence**

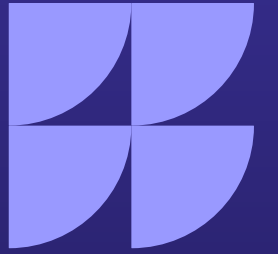
Remas Rawas – S24109161

Reema Abu Alnaja – S24209805

Ward Shaker – S24109277

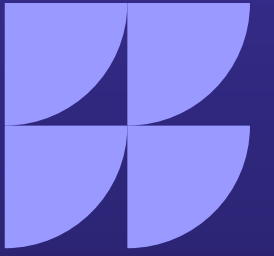
Rimas Awassabi – S24109600

Task 1.1: Knowledge Base



The program prints True because rain is true in every model that satisfies the knowledge base (KB). Only one row satisfies the KB, and rain is true in that row. Therefore, the KB entails rain.

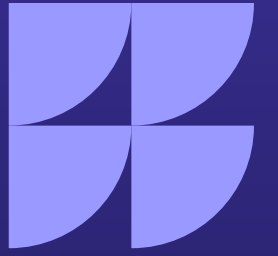
Rain	Hagrid	Dumbledore	KB true
True	True	True	False
True	True	False	False
True	False	True	True
True	False	False	False
False	True	True	False
False	True	False	False
False	False	True	False
False	False	False	False



Task 1.2: Clue Puzzle

- There must be exactly one person, one room, and one weapon.
- Ms. Scarlet: Mustard and Plum are ruled out.
- Library: Ballroom and Kitchen are ruled out.
- Knife: The initial cards rule out Revolver. The unknown card states that Scarlet, Library, and Wrench cannot all be true. Since Scarlet and Library are true, Wrench must be false.

Output: MsScarlet, Library, and Knife.

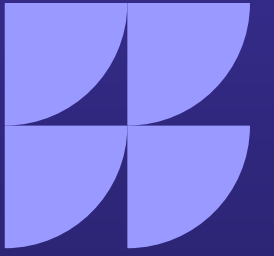


Task 2.1: Removing a Fact

- **Change:** Remove the final fact, dumbledore.
- **Prediction:** The program will print False.
- **Result:** The program printed False.
- **Explanation:** Without this fact, the KB no longer entails rain. This means rain is not guaranteed by the remaining facts.

Task 2.2: Removing a Clue

- **Change:** Remove the clue MinervaGryffindor.
- **Prediction:** Multiple house assignments will be possible.
- **Result:** Eight valid assignments remain, and the program prints no house assignments.
- **Explanation:** No person–house assignment is true in every valid model, so the KB cannot determine anyone's house uniquely.

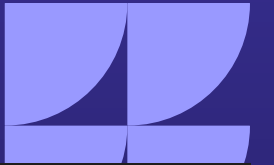


Task 2.2: Adding a New Clue

New clue: `knowledge.add(Symbol("MinervaRavenclaw"))`

- **Reported output:**
- **Gilderoy → Gryffindor**
- **Pomona → Hufflepuff**
- **Minerva → Ravenclaw**
- **Horace → Slytherin**
- **Explanation:** The new clue places Minerva in Ravenclaw. Together with the remaining constraints, it determines a unique house assignment for everyone.

Task 3



```
CKATHON
__pycache__
blue.py
harry.py
knights.py
logic.py
mastermind.py
puzzle.py

knights.py > ...
1 from logic import *
2
3 AKnight = Symbol("A is a Knight")
4 AKnave = Symbol("A is a Knave")
5 BKnight = Symbol("B is a Knight")
6 BKnave = Symbol("B is a Knave")
7
8 knowledge0 = And(
9     Or(AKnight, AKnave),
10    Not(And(AKnight, AKnave)),
11    Implication(AKnight, And(AKnight, AKnave)),
12    Implication(AKnave, Not(And(AKnight, AKnave)))
13 )
14
15 knowledge1 = And(
16     Or(AKnight, AKnave), Not(And(AKnight, AKnave)),
17     Or(BKnight, BKnave), Not(And(BKnight, BKnave)),
18     Implication(AKnight, And(AKnave, BKnave)),
19     Implication(AKnave, Not(And(AKnave, BKnave)))
20 )
21
22
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PomonaHufflepuff
MinervaGryffindor
HoraceSlytherin
● wardshakir@wz hackathon % cd /Users/wardshakir/Downloads/AI/hackathon ; /usr/bin/env /usr/local/bin/python3 /Users/wardshakir/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapters/.../debugpy/launcher 57424 -- /Users/wardshakir/Downloads/AI/hackathon/puzzle.py
● wardshakir@wz hackathon % cd /Users/wardshakir/Downloads/AI/hackathon ; /usr/bin/env /usr/local/bin/python3 /Users/wardshakir/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapters/.../debugpy/launcher 57468 -- /Users/wardshakir/Downloads/AI/hackathon/puzzle.py
● wardshakir@wz hackathon % python3 knights.py
Puzzle 0
A is a Knave
Puzzle 1
A is a Knave
B is a Knight
Puzzle 2
A is a Knave
B is a Knight
● wardshakir@wz hackathon %
```

TLINE
IELINE

```
Welcome  Untitled-1  harry.py  knights.py

harry.py > ...
1 from logic import *
2
3 rain = Symbol("rain")
4 hagrid = Symbol("hagrid")
5 dumbledore = Symbol("dumbledore")
6
7 knowledge = And(
8     Implication(Not(rain), hagrid),
9     Or(hagrid, dumbledore),
10 )
11
12
13 print(model_check(knowledge, rain))
14
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
/usr/bin/env /usr/local/bin/python3 /Users/wardshakir/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapters/.../debugpy/launcher 57303 -- /Users/wardshakir/Downloads/AI/hackathon/harry.py
● wardshakir@wz hackathon % /usr/bin/env /usr/local/bin/python3 /Users/wardshakir/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapters/.../debugpy/launcher 57303 -- /Users/wardshakir/Downloads/AI/hackathon/harry.py
True
● wardshakir@wz hackathon % cd /Users/wardshakir/Downloads/AI/hackathon ; /usr/bin/env /usr/local/bin/python3 /Users/wardshakir/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapters/.../debugpy/launcher 57325 -- /Users/wardshakir/Downloads/AI/hackathon/harry.py
False
○ wardshakir@wz hackathon %
```

```
puzzle.py > ...
1 from logic import *
2
3 people = ["Gilderoy", "Pomona", "Minerva", "Horace"]
4 houses = ["Gryffindor", "Hufflepuff", "Ravenclaw", "Slytherin"]
5
6 symbols = []
7
8 knowledge = And()
9
10 for person in people:
11     for house in houses:
12         symbols.append(Symbol(f"{person}{house}"))
13
14 # Each person belongs to a house.
15 for person in people:
16     knowledge.add(Or(
17         Symbol(f"{person}Gryffindor"),
18         Symbol(f"{person}Hufflepuff"),
19         Symbol(f"{person}Ravenclaw"),
20         Symbol(f"{person}Slytherin")
21     ))
22
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
/usr/bin/env /usr/local/bin/python3 /Users/wardshakir/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapters/.../debugpy/launcher 57303 -- /Users/wardshakir/Downloads/AI/hackathon/harry.py
True
● wardshakir@wz hackathon % cd /Users/wardshakir/Downloads/AI/hackathon ; /usr/bin/env /usr/local/bin/python3 /Users/wardshakir/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapters/.../debugpy/launcher 57325 -- /Users/wardshakir/Downloads/AI/hackathon/harry.py
False
● wardshakir@wz hackathon % cd /Users/wardshakir/Downloads/AI/hackathon ; /usr/bin/env /usr/local/bin/python3 /Users/wardshakir/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapters/.../debugpy/launcher 57378 -- /Users/wardshakir/Downloads/AI/hackathon/puzzle.py
GilderoyRavenclaw
PomonaHufflepuff
MinervaGryffindor
HoraceSlytherin
○ wardshakir@wz hackathon %
```

3

```
● wardshakir@wz lab-3-AI % python3 -m pip install termcolor
Collecting termcolor
  Downloading termcolor-3.3.0-py3-none-any.whl.metadata (6.5 kB)
  Downloading termcolor-3.3.0-py3-none-any.whl (7.7 kB)
Installing collected packages: termcolor
Successfully installed termcolor-3.3.0
```

```
[notice] A new release of pip is available: 24.2 -> 26.2.1
[notice] To update, run: pip3 install --upgrade pip
```

```
● wardshakir@wz lab-3-AI % python3 harry.py
True
○ wardshakir@wz lab-3-AI %
```